Reg. No. | | | | | | | | | |

Question Paper Code : 12659

B.E / B.TECH DEGREE EXAMINATION, NOV / DEC 2016

Third Semester

Common to B.E – CSE & B.Tech - IT

15CS302 – DIGITAL PRINCIPLES AND SYSTEM DESIGN

(Regulations: Mepco – R2015)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

PART A – (10 × 2 = 20 Marks)

1. Find the minterm expansion of f(a,b,c,d)= a'(b'+d)+acd'.
2. State and prove consensus theorem.
3. Why NAND and NOR gates are called universal gates?
4. Draw the block representation of 4 bit binary parallel adder.
5. List the types of ROM.
6. What is multiplexed Display systems?
7. What is race around condition? How is it avoided?
8. How will you convert D Flip-flop into T Flip-flop?
9. How an essential hazard can be eliminated?
10. What is asynchronous sequential circuit?

PART B – (5 × 16 = 80 Marks)

11. a) i. Simplify the Boolean expressions to a minimum number of literals.

A. $xy + xy'$

B. $xy + x(wz + wz')$

C. $(BC'+A'D)(AB'+CD')$ (8 Marks)

11. a) ii. Find a minimal sum of products representation for F(A,B,C,D,E)= $\sum_m$ (1,4,6,10,20,22,24,26) + d(0,11,16,27) using Karnaugh map. (8 Marks)

OR

11. b) Find a minimum sum of products expression for the following function using Quine-Mc Cluskey method
$f(A,B,C,D,E)=\sum(0,2,3,5,7,9,11,13,14,16,18,24,26,28,30)$ (16 Marks)

12. a) i. A system of logic is to be designed which has two outputs and three inputs. One output will be TRUE if an odd number of inputs are TRUE. The other output will be TRUE if only one input is alone TRUE. Draw the truth table and write the corresponding Boolean equations. (6 Marks)

12. a) ii. Design a BCD to Gray Code Converter. Use don't cares. (10 Marks)

OR

12. b) i. Realize $F(w,x,y,z) = \sum_m(1,4,6,7,8,9,10,11,15)$ using 8X1 MUX. (8 Marks)

12. b) ii. Write a gate level HDL description for 3X8 decoder. (8 Marks)

13. a) i. Implement the switching functions :

A. Z_1= ab'd'e+ a'b'c'd'e'+bc+de
B. Z_2= a'c'e
C. Z_3=bc+de+c'd'e'+bd
D. Z_4=a'c'e+ce

Using 5 x 8 x 4 PLA (10 Marks)

13. a) ii. Discuss about Error detection and correction with suitable examples. (6 Marks)

OR

13. b) i. Discuss about Multiplexed display systems. (8 Marks)

13. b) ii Using a ROM, implement a combinatorial circuit which accepts 3 bit number and generates an output binary number equal to the square of the input number. (8 Marks)

14. a) Using Positive edge triggering SR flip – flops design a counter which counts in the following sequence: 000,111,110,101,100,011,010,001,000,……. (16 Marks)

OR

14. b) i. A Sequential circuit with Two D FFs A and B and input X and output Y is specified by the following next state and output equations.

$$A(t+1) = AX + BX$$

$$B(t+1) = A'X$$

$$Y = (A+B)X'$$

A. Draw the logic diagram of the circuit

B. Derive the state table

C. Derive the state diagram (10 Marks)

14. b) ii. Write the HDL description for Asynchronous BCD Counter. (6 Marks)

15. a) i. An asynchronous sequential circuit is described by the following excitation and output function

$$Y = X_1X_2 + (X_1+X_2)\, y$$

$$Z = y$$

A. Draw the logic diagram of the circuit.

B. Derive the transition table, flow table

C. Derive the output map. (10 Marks)

15. a) ii. Write the analysis procedure for analyzing an asynchronous sequential circuit. (6 Marks)

OR

15. b) i. Discuss about various hazards. (8 Marks)

15. b) ii. Give hazard free realization for the following Boolean function $F(a,b,c,d)= \Sigma m(1,3,6,7,13,15)$. (8 Marks)